



Home <<https://epa.gov/>> / Energy and the Environment <<https://epa.gov/energy>>

Greenhouse Gas Equivalencies Calculator – Calculations and References

This page describes the calculations used to convert greenhouse gas emission numbers into different types of equivalent units. Go to the Greenhouse Gas Equivalencies Calculator page for more information <<https://epa.gov/energy/greenhouse-gas-equivalencies-calculator>>.

A note on global warming potentials (GWPs): Some of the equivalencies in the calculator are reported as CO₂ equivalents (CO₂e). These are calculated using GWPs from the Intergovernmental Panel on Climate Change's Fifth Assessment Report (AR5), except for the equivalencies that are calculated using the Waste Reduction Model (WARM) because the model uses the Fourth Assessment Report (tons of waste recycled instead of landfilled, GHG emissions avoided by recycling X garbage trucks or railcars full of waste, and trash bags of waste recycled instead of landfilled). To facilitate user reproducibility calculations that use data from WARM, the input data were not modified to reflect GWPs from AR5.

eGRID Regions

As of 2024, the Greenhouse Gas Equivalencies Calculator uses the Emissions & Generation Resource Integrated Database (eGRID), which derives regional emission factors for electric power marginal and baseload emissions and enables users to map their residential or commercial zip code to their region. This results in more accurate

emissions estimates regarding their energy use (relative to the previous national default) and affects the electricity used and avoided factors. The regionally specific and national weighted emission rates are presented in Table 1 .

Table 1 : Emissions & Generation Resource Integrated Database (eGRID) Subregion Output Emission Rates (eGRID2022)

eGRID subregion acronym	eGRID subregion name	Total output (baseload) emission rate (lb CO₂/mWh)	Non-baseload (marginal) output emission rate (lb CO₂/mWh)
AKGD	ASCC Alaska Grid	1,052.1	1,224.5
AKMS	ASCC Miscellaneous	495.8	1,587.9
AZNM	WECC Southwest	776.0	1,205.2
CAMX	WECC California	497.4	1,055.0
ERCT	ERCOT All	771.1	1,194.9
FRCC	FRCC All	813.8	1,044.4
HIMS	HICC Miscellaneous	1,155.5	1,619.2
HIOA	HICC Oahu	1,575.4	1,810.3
MROE	MRO East	1,479.6	1,672.9
MROW	MRO West	936.5	1,794.7
NEWE	NPCC New England	536.4	923.3
NWPP	WECC Northwest	602.1	1,515.7

eGRID subregion acronym	eGRID subregion name	Total output (baseload) emission rate (lb CO₂/mWh)	Non-baseload (marginal) output emission rate (lb CO₂/mWh)
NYCW	NPCC NYC/Westchester	885.2	971.8
NYLI	NPCC Long Island	1,200.7	1,316.7
NYUP	NPCC Upstate NY	274.6	920.1
PRMS	Puerto Rico Miscellaneous	1,593.5	1,670.9
RFCE	RFC East	657.4	1,278.7
RFCM	RFC Michigan	1,216.4	1,597.3
RFCW	RFC West	1,000.1	1,843.6
RMPA	WECC Rockies	1,124.9	1,676.4
SPNO	SPP North	952.6	1,943.0
SPSO	SPP South	970.4	1,528.2
SRMV	SERC Mississippi Valley	801.0	1,220.7
SRMW	SERC Midwest	1,369.9	1,808.6
SRSO	SERC South	893.3	1,354.8
SRTV	SERC Tennessee Valley	933.1	1,671.0
SRVC	SERC Virginia/Carolina	623.0	1,308.8

eGRID subregion acronym	eGRID subregion name	Total output (baseload) emission rate (lb CO ₂ /mWh)	Non-baseload (marginal) output emission rate (lb CO ₂ /mWh)
U.S.		823.1	1,405.3

Notes

- The zip code box will only appear if the user selects the emissions avoided or reduced data input.
- If no zip code is entered by the user, the national default will be used (which was used prior to the 2024 update).
- If a non-existent zip code is used, a warning sign will be displayed.

Sources

- EPA (2024a). eGRID <<https://epa.gov/egrid>>. U.S. marginal and baseload emission rates, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.
- EPA (2024b). Power Profiler, year 2022 eGRID data. U.S. Environmental Protection Agency, Washington, DC.

Electricity avoided (kilowatt-hours)

The Greenhouse Gas Equivalencies Calculator uses the Emissions & Generation Resource Integrated Database (eGRID) regionally specific or national weighted average CO₂ marginal emission rate to convert avoided kilowatt-hours into avoided units of carbon dioxide emissions, depending on the user input.

Most users of the Equivalencies Calculator who seek equivalencies for electricity-related emissions want to know equivalencies for emissions avoided from energy efficiency (EE) or renewable energy (RE) programs. Calculating the emission impacts of EE and RE on the electricity grid requires estimating the amount of fossil-fired generation and emissions being displaced by EE and RE. A marginal emission factor is the best representation to estimate which fossil-fired units EE/RE are displacing across the fossil fleet. EE and RE programs are not generally assumed to affect baseload power

plants that run all the time, but rather marginal power plants that are brought online as necessary to meet demand. Therefore, eGRID provides either a regionally specific *or* national marginal emission factor for the Equivalencies Calculator, depending on the user input.

Emission Factor

$1,405.3 \text{ lbs CO}_2/\text{MWh} \times 1 \text{ metric ton}/2,204.6 \text{ lbs} \times 1/(1-0.051) \text{ MWh delivered}/\text{MWh generated} \times 1 \text{ MWh}/1,000 \text{ kWh} = 6.72 \times 10^{-4} \text{ metric tons CO}_2/\text{kWh}$

(eGRID, U.S. national weighted average CO₂ marginal emission rate, year 2022 data)

Notes:

- The calculation shown here utilizes the national default marginal emission rate. Table 1 shows the eGRID regionally specific and national marginal emission rates.
- This calculation does not include any greenhouse gases other than CO₂.
- This calculation includes line losses.

Sources

- EPA (2024a). eGRID <<https://epa.gov/egrid>>. U.S. marginal and baseload emission rates, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.

Electricity used (kilowatt-hours)

The Greenhouse Gas Equivalencies Calculator uses the eGRID U.S. regionally specific *or* national annual average CO₂ output rate to convert kilowatt-hours of energy use into units of carbon dioxide emissions, depending on the user input.

This calculation is intended for users who would like to know the equivalencies associated with greenhouse gas emissions associated with electricity used, not avoided. This is either a regionally specific *or* a national average emissions factor, depending on the user input.

Emission Factor

$823.1 \text{ lbs CO}_2/\text{MWh} \times 1 \text{ metric ton}/2,204.6 \text{ lbs} \times 1/(1-0.051) \text{ MWh delivered/MWh generated} \times 1 \text{ MWh}/1,000 \text{ kWh} = 3.94 \times 10^{-4} \text{ metric tons CO}_2/\text{kWh}$

(eGRID, U.S. annual CO₂ total output emission rate [lb/MWh], year 2022 data)

Notes:

- The calculation shown here utilizes the national default marginal emission rate. Table 1 shows the eGRID regionally specific and national marginal emission rates.
- This calculation includes line losses.
- Regional average emission rates are also available on the eGRID <<https://epa.gov/egrid>> web page.

Sources

- EIA (2023b). 2022 State Electricity Profiles U.S. Total, Table 10: Supply and Disposition of Electricity
- EPA (2024). eGRID <<https://epa.gov/egrid>>. Table 1: Subregion Output Emission Rates (eGRID2022), year 2022 data. U.S. Environmental Protection Agency, Washington, DC.

Gallons of gasoline consumed

In the preamble to the joint EPA/Department of Transportation rulemaking on May 7, 2010 that established the initial National Program fuel economy standards for model years 2012-2016, the agencies stated that they had agreed to use a common conversion factor of 8,887 grams of CO₂ emissions per gallon of gasoline consumed (Federal Register 2010). For reference, to obtain the number of grams of CO₂ emitted per gallon of gasoline combusted, the heat content of the fuel per gallon can be multiplied by the kg CO₂ per heat content of the fuel.

This value assumes that all the carbon in the gasoline is converted to CO₂ (IPCC 2006).

Calculation

8,887 grams of CO₂/gallon of gasoline = 8.887×10^{-3} metric tons CO₂/gallon of gasoline

Sources

- Federal Register (2010). Light-Duty Vehicle Greenhouse Gas Emission Standards and Corporate Average Fuel Economy Standards; Final Rule, page 25,330 (PDF) [⌕](http://www.gpo.gov/fdsys/pkg/fr-2010-05-07/pdf/2010-8159.pdf)
<<http://www.gpo.gov/fdsys/pkg/fr-2010-05-07/pdf/2010-8159.pdf>> (407 pp, 5.7MB, About PDF
<<https://epa.gov/home/pdf-files>>).
- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland [⌕](https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html)
<<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>>.

Gallons of diesel consumed

In the preamble to the joint EPA/Department of Transportation rulemaking on May 7, 2010 that established the initial National Program fuel economy standards for model years 2012-2016, the agencies stated that they had agreed to use a common conversion factor of 10,180 grams of CO₂ emissions per gallon of diesel consumed (Federal Register 2010). For reference, to obtain the number of grams of CO₂ emitted per gallon of diesel combusted, the heat content of the fuel per gallon can be multiplied by the kg CO₂ per heat content of the fuel.

This value assumes that all the carbon in the diesel is converted to CO₂ (IPCC 2006).

Calculation

10,180 grams of CO₂/gallon of diesel = 10.180×10^{-3} metric tons CO₂/gallon of diesel

Sources

- Federal Register (2010). Light-Duty Vehicle Greenhouse Gas Emission Standards and Corporate Average Fuel Economy Standards; Final Rule, page 25,330 (PDF) [⌵](http://www.gpo.gov/fdsys/pkg/fr-2010-05-07/pdf/2010-8159.pdf)
<<http://www.gpo.gov/fdsys/pkg/fr-2010-05-07/pdf/2010-8159.pdf>> (407 pp, 5.7MB, About PDF
<<https://epa.gov/home/pdf-files>>).
- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland [⌵](https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html)
<<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>>.

Gasoline-powered passenger vehicles per year

Passenger vehicles are defined as 2-axle 4-tire vehicles, including passenger cars, vans, pickup trucks, and sport/utility vehicles.

In 2022, the weighted average combined fuel economy of cars and light trucks was 22.8 miles per gallon (FHWA 2024). The average vehicle miles traveled (VMT) in 2022 was 10,917 miles per year (FHWA 2024).

In 2022, the ratio of carbon dioxide emissions to total greenhouse gas emissions (including carbon dioxide, methane, and nitrous oxide, all expressed as carbon dioxide equivalents) for gasoline passenger vehicles was 0.994 (EPA 2024b).

The amount of carbon dioxide emitted per gallon of motor gasoline burned is 8.89×10^{-3} metric tons, as calculated in the “Gallons of gasoline consumed” section above.

To determine annual greenhouse gas emissions per passenger vehicle, the following methodology was used: VMT was divided by average gas mileage to determine gallons of gasoline consumed per vehicle per year. Gallons of gasoline consumed was multiplied by carbon dioxide per gallon of gasoline to determine carbon dioxide emitted per vehicle per year. Carbon dioxide emissions were then divided by the ratio of carbon dioxide emissions to total vehicle greenhouse gas emissions to account for vehicle methane and nitrous oxide emissions.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

8.89×10^{-3} metric tons CO₂/gallon gasoline $\times$ 10,917 VMT_{car/truck average} $\times$ 1/22.8 miles per gallon_{car/truck average} $\times$ 1 CO₂, CH₄, and N₂O/0.994 CO₂ = **4.29 metric tons CO₂e/vehicle /year**

Sources

- EPA (2024b). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Chapter 3 (Energy), Tables 3-13, 3-14, and 3-15
<<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-chapter-3-energy.pdf>>. Pg. 3-27 to 3-31, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>)
- FHWA (2024). Highway Statistics 2022 [🔗](https://www.fhwa.dot.gov/policyinformation/statistics/2022/vm1.cfm)
<<https://www.fhwa.dot.gov/policyinformation/statistics/2022/vm1.cfm>>. Office of Highway Policy Information, Federal Highway Administration. Table VM-1. (1 pp, 92 KB, About PDF <<https://epa.gov/home/pdf-files>>)

Electric passenger vehicles per year

Electric passenger vehicles are defined as all-electric vehicles that are full-sized, sold in the United States, and are capable of achieving a speed of 60 mph.

The weighted average combined electric efficiency of U.S. electric vehicles sales from 2019 and earlier is 3.60 miles per kWh (DOE 2023). The average vehicle miles traveled (VMT) in 2022 was 10,917 miles per year and is assumed to be the same for electric cars (FHWA 2024). The amount of carbon dioxide equivalent emitted per MWh is 823.1 pounds (EPA 2024).

To determine annual greenhouse gas emissions per all-electric passenger vehicle, the following methodology was used: VMT was divided by the average miles per kWh to determine kWh consumed per vehicle per year. Then, kWh consumed were converted to MWh consumed by multiplying the ratio of MWh to kWh. Electricity consumed was multiplied by pounds of carbon dioxide equivalent per MWh to determine carbon

dioxide equivalent emitted per vehicle per year. Carbon dioxide emissions were then converted from pounds to metric tons by multiplying by the ratio of metric tons to pounds.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$823.1 \text{ lbs CO}_2\text{e/MWh} \times 10,917 \text{ VMT}_{\text{car/truck average}} \times 1/3.60 \text{ miles per kWh all EVs average} \times 1/1000 \text{ MWh/kWh} \times 1 \text{ metric ton}/2,204.6 \text{ lbs} = \mathbf{1.13 \text{ metric tons CO}_2\text{e/vehicle/year}}$

Sources

- DOE (2023). Alternative Fuels Data Center: Data Sources and Assumptions for the Electricity Sources and Fuel-Cycle Emissions Tool.
- EPA (2024). eGRID <<https://epa.gov/egrid/detailed-data>>, U.S. annual national emission factor, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.
- FHWA (2024). Highway Statistics 2022 [🔗](https://www.fhwa.dot.gov/policyinformation/statistics/2022/vm1.cfm)
<<https://www.fhwa.dot.gov/policyinformation/statistics/2022/vm1.cfm>>. Office of Highway Policy Information, Federal Highway Administration. Table VM-1. (1 pp, 92 KB, About PDF
<<https://epa.gov/home/pdf-files>>)

Miles driven by the average gasoline-powered passenger vehicle

Passenger vehicles are defined as 2-axle 4-tire vehicles, including passenger cars, vans, pickup trucks, and sport/utility vehicles.

In 2022, the weighted average combined fuel economy of cars and light trucks was 22.8 miles per gallon (FHWA 2024). In 2022, the ratio of carbon dioxide emissions to total greenhouse gas emissions (including carbon dioxide, methane, and nitrous oxide, all expressed as carbon dioxide equivalents) for passenger vehicles was 0.994 (EPA 2024).

The amount of carbon dioxide emitted per gallon of motor gasoline burned is 8.89×10^{-3} metric tons, as calculated in the “Gallons of gasoline consumed” section above.

To determine annual greenhouse gas emissions per mile, the following methodology was used: carbon dioxide emissions per gallon of gasoline were divided by the average fuel economy of vehicles to determine carbon dioxide emitted per mile traveled by a typical passenger vehicle. Carbon dioxide emissions were then divided by the ratio of carbon dioxide emissions to total vehicle greenhouse gas emissions to account for vehicle methane and nitrous oxide emissions.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$$8.89 \times 10^{-3} \text{ metric tons CO}_2/\text{gallon gasoline} \times 1/22.8 \text{ miles per gallon}_{\text{car/truck average}} \times 1 \text{ CO}_2, \text{ CH}_4, \text{ and N}_2\text{O}/0.994 \text{ CO}_2 = \mathbf{3.93 \times 10^{-4} \text{ metric tons CO}_2\text{e/mile}}$$

Sources

- EPA (2024). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Chapter 3 (Energy), Tables 3-13, 3-14, and 3-15
<<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-chapter-3-energy.pdf>>. Pg. 3-27 to 3-31, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>)
- FHWA (2024). Highway Statistics 2024. Office of Highway Policy Information, Federal Highway Administration. Table VM-1 [↗](#)
<<https://www.fhwa.dot.gov/policyinformation/statistics/2022/vm1.cfm>>. (1 pp, 193 KB, About PDF <<https://epa.gov/home/pdf-files>>)

Therms and Mcf of natural gas

Carbon dioxide emissions per therm are determined by converting million British thermal units (mmbtu) to therms, then multiplying the carbon coefficient times the fraction oxidized times the ratio of the molecular weight of carbon dioxide to carbon (44/12).

0.1 mmbtu equals one therm (EIA 2024). The average carbon coefficient of pipeline natural gas burned in 2022 is 14.43 MMT C/QBtu (EPA 2024). The fraction oxidized to CO₂ is assumed to be 100 percent (IPCC 2006).

Note: When using this equivalency, please keep in mind that it represents the CO₂ equivalency of CO₂ released for natural gas **burned** as a fuel, not natural gas released to the atmosphere. Direct methane emissions released to the atmosphere (without burning) are about 28 times more powerful than CO₂ in terms of their warming effect on the atmosphere.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$$0.1 \text{ mmbtu}/1 \text{ therm} \times 14.43 \text{ kg C/mmbtu} \times 44 \text{ kg CO}_2/12 \text{ kg C} \times 1 \text{ metric ton}/1,000 \text{ kg} = \mathbf{0.0053 \text{ metric tons CO}_2/\text{therm}}$$

Carbon dioxide emissions per therm can be converted to carbon dioxide emissions per thousand cubic feet (Mcf) using the average heat content of natural gas in 2022, 10.360 therms/Mcf (EIA 2024).

$$0.0053 \text{ metric tons CO}_2/\text{therm} \times 10.40 \text{ therms/Mcf} = \mathbf{0.0548 \text{ metric tons CO}_2/\text{Mcf}}$$

Sources

- EIA (2024). Heat Content of Natural Gas Consumed. [🔗](https://www.eia.gov/dnav/ng/ng_cons_heat_a_epg0_vgth_btucf_a.htm)
<https://www.eia.gov/dnav/ng/ng_cons_heat_a_epg0_vgth_btucf_a.htm> (PDF) (1 pp, 54 KB, About PDF <<https://epa.gov/home/pdf-files>>)
- EIA (2023). Natural Gas Conversions – Frequently Asked Questions [🔗](#).
- EPA (2024). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Annex 2 (Methodology for estimating CO₂ emissions from fossil fuel combustion) <<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-annex-2-emissions-fossil-fuel-combustion.pdf>>, Table A-20 “C Content Coefficients by Year (MMT C/QBtu)” pg. A-51, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>)

- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland [✉](http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html)
<<http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>>.

Barrels of oil consumed

Carbon dioxide emissions per barrel of crude oil are determined by multiplying heat content times the carbon coefficient times the fraction oxidized times the ratio of the molecular weight of carbon dioxide to that of carbon (44/12).

The average heat content of crude oil is 5.80 mmbtu per barrel (EPA 2024). The average carbon coefficient of crude oil is 20.31 MMT C/QBtu (EPA 2024). The fraction oxidized is assumed to be 100 percent (IPCC 2006).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$5.80 \text{ mmbtu/barrel} \times 20.31 \text{ kg C/mmbtu} \times 44 \text{ kg CO}_2/12 \text{ kg C} \times 1 \text{ metric ton}/1,000 \text{ kg} =$
0.43 metric tons CO₂/barrel

Sources

- EPA (2023). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Annex 2 (Methodology for estimating CO₂ emissions from fossil fuel combustion) <<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-annex-2-emissions-fossil-fuel-combustion.pdf>>, Table A-20 "C Content Coefficients by Year (MMT C/QBtu)," pg. A-51, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (760 pp, 15.9 MB, About PDF)
- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland [✉](http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html)
<[https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html](http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html)>.

Tanker trucks filled with gasoline

The amount of carbon dioxide emitted per gallon of motor gasoline burned is 8.89×10^{-3} metric tons, as calculated in the “Gallons of gasoline consumed” section above. A barrel equals 42 gallons. A typical gasoline tanker truck contains 8,500 gallons.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

8.89×10^{-3} metric tons CO₂/gallon $\times$ 8,500 gallons/tanker truck = **75.54 metric tons CO₂/tanker truck**

Sources

- Federal Register (2010). Light-Duty Vehicle Greenhouse Gas Emission Standards and Corporate Average Fuel Economy Standards; Final Rule, page 25,330 (PDF) <http://www.gpo.gov/fdsys/pkg/fr-2010-05-07/pdf/2010-8159.pdf> (407 pp, 5.7MB, About PDF <https://epa.gov/home/pdf-files>).
- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland <https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>.

Home electricity use

In 2022, 126.4 million homes in the United States consumed 1,541 billion kilowatt-hours (kWh) of electricity (EIA 2023a). On average, each home consumed 12,194 kWh of delivered electricity (EIA 2023a). The national average carbon dioxide output rate for electricity generated in 2022 was 823.1 lbs CO₂ per megawatt-hour (EPA 2024a), which translates to about 867.5 lbs CO₂ per megawatt-hour for delivered electricity, assuming transmission and distribution losses of 5.12% (EIA 2023b; EPA 2024b).

The annual 2022 U.S. transmission and distribution (T&D) losses were determined as estimated losses divided by the total disposition less direct use. This T&D loss percentage considers utility-reported data including estimated T&D losses that occur

for the electricity transmitted and distributed in the United States based on total disposition (i.e., total electricity supply) net of direct use (i.e., electricity used by the same industrial or commercial sector facility where it is produced, thus never entering the grid).

Annual home electricity consumption was multiplied by the carbon dioxide emission rate (per unit of electricity delivered) to determine annual carbon dioxide emissions per home.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$212.4 \text{ billion kWh} / (4,287.6 \text{ billion kWh} - 139.7 \text{ billion kWh}) = 5.12\% \text{ MWh lost from T\&D}$

$12,194 \text{ kWh per home} \times 823.1 \text{ lbs CO}_2 \text{ per megawatt-hour generated} \times 1/(1-0.051) \text{ MWh delivered/MWh generated} \times 1 \text{ MWh}/1,000 \text{ kWh} \times 1 \text{ metric ton}/2,204.6 \text{ lb} = \mathbf{4.798 \text{ metric tons CO}_2/\text{home}}.$

Sources

- EIA (2023a). 2023 Annual Energy Outlook, Table 4: Residential Sector Key Indicators and Consumption [↗](#).
- EIA (2023b) 2022 State Electricity Profiles U.S. Total, Table 10: Supply and Disposition of Electricity.
- EPA (2024a). eGRID <https://epa.gov/egrid>, U.S. annual national emission factor, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.

Home energy use

In 2022, there were 126.4 million homes in the United States (EIA 2023a). On average, each home consumed 12,194 kWh of delivered electricity. Nationwide household consumption of natural gas, propane, and fuel oil totaled 5.15, 0.49, and 0.43 quadrillion Btu, respectively, in 2022 (EIA 2023a). Averaged across households in the United States, this amounts to 39,319 cubic feet of natural gas, 42.5 gallons of propane, and 24.7 gallons of fuel oil per home.

The national average carbon dioxide output rate for generated electricity in 2021 was 823.1 lbs CO₂ per megawatt-hour (EPA 2024a), which translates to about 867.5 lbs CO₂ per megawatt-hour for delivered electricity (assuming transmission and distribution losses of 5.12%) (EPA 2024a; EIA 2023b).¹ <<https://epa.gov/energy/greenhouse-gases-equivalencies-calculator-calculations-and-references#footnote1>>

The average carbon dioxide coefficient of natural gas is 14.43 MMT C/QBtu, which converts to 0.0550 kg CO₂ per cubic foot (EPA 2024b). The fraction oxidized to CO₂ is 100 percent (IPCC 2006).

The average carbon dioxide coefficient of distillate fuel oil is 20.22 MMT C/QBtu, which converts to 431.87 kg CO₂ per 42-gallon barrel (EPA 2024b). The fraction oxidized to CO₂ is 100 percent (IPCC 2006).

The gross heat content of LPG is 3.841 MMBtu/Barrel, which converts to an average carbon dioxide coefficient of 236.0 kg CO₂ per 42-gallon barrel (EPA 2024c). The fraction oxidized is 100 percent (IPCC 2006).

Total home electricity, natural gas, distillate fuel oil, and propane consumption figures were converted from their various units to metric tons of CO₂ and added together to obtain total CO₂ emissions per home.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

1. Electricity: 12,194 kWh per home × 823.1 lbs CO₂ per megawatt-hour generated × (1/(1-0.051)) MWh generated/MWh delivered × 1 MWh/1,000 kWh × 1 metric ton/2,204.6 lbs = 4.798 metric tons CO₂/home.

2. Natural gas: 39,319 cubic feet per home × 0.0550 kg CO₂/cubic foot × 1/1,000 kg/metric ton = 2.16 metric tons CO₂/home

3. Propane: 42.5 gallons per home × 1/42 barrels/gallon × 236.0 kg CO₂/barrel × 1/1,000 kg/metric ton = 0.24 metric tons CO₂/home

4. Fuel oil: 24.7 gallons per home × 1/42 barrels/gallon × 431.87 kg CO₂/barrel × 1/1,000 kg/metric ton = 0.25 metric tons CO₂/home

Total CO₂ emissions for energy use per home: 4.798 metric tons CO₂ for electricity + 2.16 metric tons CO₂ for natural gas + 0.24 metric tons CO₂ for propane + 0.25 metric tons CO₂ for fuel oil = **7.45 metric tons CO₂ per home per year.**

Sources

- EIA (2023a). 2023 Annual Energy Outlook, Table 4: Residential Sector Key Indicators and Consumption [↗](#).
- EIA (2023b) 2022 State Electricity Profiles U.S. Total, Table 10: Supply and Disposition of Electricity.
- EIA (2024). Heat Content of Natural Gas Delivered to Consumers [↗](#)
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- EPA (2024a). eGRID <<https://epa.gov/egrid>>, U.S. annual national emission factor, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.
- EPA (2024b). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Annex 2 (Methodology for estimating CO₂ emissions from fossil fuel combustion) <<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-annex-2-emissions-fossil-fuel-combustion.pdf>>, Table A-20 "C Content Coefficients by Year (MMT C/QBtu)," pg. A-51, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>)
- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland [↗](#)
<<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>>.

Number of urban tree seedlings grown for 10 years

A medium growth coniferous or deciduous tree, planted in an urban setting and allowed to grow for 10 years, sequesters 23.2 and 38.0 lbs of carbon, per year respectively. These estimates are based on the following assumptions:

- The medium growth coniferous and deciduous trees are raised in a nursery for one year until they become 1 inch in diameter at 4.5 feet above the ground (the size of tree purchased in a 15-gallon container).
- The nursery-grown trees are then planted in a suburban/urban setting; the trees are not densely planted.
- The calculation takes into account “survival factors” developed by U.S. DOE (1998). For example, after 5 years (one year in the nursery and 4 in the urban setting), the probability of survival is 68 percent; after 10 years, the probability declines to 59 percent. To estimate losses of growing trees, in lieu of a census conducted to accurately account for the total amount of seedlings planted versus surviving to a certain age, the sequestration rate (in lbs per tree per year) is multiplied by the survival factor to yield a probability-weighted annual sequestration rate. These values are summed for the 10-year period, beginning from the time of planting, and divided by the 10-year period, to derive the annual average sequestration estimate of 23.2 lbs of carbon per coniferous tree or 38.0 lbs of carbon per deciduous tree.

The estimates of carbon sequestered by coniferous and deciduous trees were then weighted by the percent share of coniferous versus deciduous trees in cities across the United States. Of a sample of approximately 14,000 coniferous and deciduous trees in seventeen major U.S. cities, approximately 11 percent and 89 percent of sampled trees were coniferous and deciduous, respectively (McPherson et al. 2016). Therefore, the weighted average annual carbon sequestered by a medium growth coniferous or deciduous tree, planted in an urban setting and allowed to grow for 10 years, is 36.4 lbs of carbon per tree. Trees sequester significantly less carbon in early life than mid life, and carbon sequestration slows in late life. The typical age of each tree species is variable as is the rate of growth, hence this method determines an annual average assuming ten years of growth.

Please note the following caveats to these assumptions:

- While most trees take 1 year in a nursery to reach the seedling stage, trees grown under different conditions and trees of certain species may take longer: up to 6 years.
- Average survival rates in urban areas are based on broad assumptions, and the rates will vary significantly depending upon site conditions.

- Carbon sequestration is dependent on growth rate, which varies by species, age, location and other conditions.
- This method estimates only direct sequestration of carbon, and does not include the energy savings that result from buildings being shaded by urban tree cover.
- This method is best used as an estimation for suburban/urban areas (i.e., parks, along sidewalks, yards) with highly dispersed tree plantings and is not appropriate for reforestation projects.

To convert to units of metric tons CO₂ per tree, multiply by the ratio of the molecular weight of carbon dioxide to that of carbon (44/12) and the ratio of metric tons per pound (1/2,204.6).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$(0.11 \text{ [percent of coniferous trees in sampled urban settings]} \times 23.2 \text{ lbs C/coniferous tree/year}) + (0.89 \text{ [percent of deciduous trees in sampled urban settings]} \times 38.0 \text{ lbs C/deciduous tree/year}) = 36.4 \text{ lbs C/tree/year}$

$36.4 \text{ lbs C/tree/year} \times (44 \text{ units CO}_2/12 \text{ units C}) \times 1 \text{ metric ton}/2,204.6 \text{ lbs} = \mathbf{0.060 \text{ metric ton CO}_2 \text{ per urban tree planted per year}}$

Sources

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<<https://www3.epa.gov/climatechange/downloads/method-calculating-carbon-sequestration-trees-urban-and-suburban-settings.pdf>> (16 pp, 111K, About PDF <<https://epa.gov/home/pdf-files>>)

Acres of U.S. forests sequestering CO₂ for one year

Forests are defined herein as managed forests that have been classified as forests for over 20 years (i.e., excluding forests converted to/from other land-use types). Please refer to the *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* for a discussion of the definition of U.S. forests and methodology for estimating carbon stored in U.S. forests (EPA 2024).

Growing forests accumulate and store carbon. Through the process of photosynthesis, trees remove CO₂ from the atmosphere and store it as cellulose, lignin, and other compounds. The rate of accumulation of carbon in a forested landscape is equal to overall tree growth minus removals (i.e., harvest for the production of paper and wood and tree loss from natural disturbances) minus decomposition. In most U.S. forests, growth exceeds removals and decomposition, so the amount of carbon stored nationally in forested lands is increasing overall, though at a decreasing rate.

Calculation for U.S. Forests

The *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* (EPA 2024) provides data on the net change in forest carbon stocks and forest area.

Annual Net Change in Carbon Stocks per Area in Year t = (Carbon Stocks($t+1$) - Carbon Stock t)/Area of land remaining in the same land-use category

Step 1: Determine the carbon stock change between years by subtracting carbon stocks in year t from carbon stocks in year ($t+1$). This calculation, also found in the *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* (EPA 2024), uses the USDA Forest Service estimates of carbon stocks in 2022 minus carbon stocks in 2021. (This calculation includes carbon stocks in the aboveground biomass, belowground biomass, dead wood, litter, and soil organic and mineral carbon pools. C gains attributed to harvested wood products are not included in this calculation.)

Annual Net Change in Carbon Stocks in Year 2022 = 62,320 MMT C – 62,130 MMT C = **189 MMT C**

Step 2: Determine the annual net change in carbon stocks (i.e., sequestration) per area by dividing the carbon stock change in U.S. forests from Step 1 by the total area of U.S. forests remaining in forests in year t (i.e., the area of land that did not change land-use categories between the time periods).

Applying the Step 2 calculation to data developed by the USDA Forest Service for the *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* yields a result of 221 metric tons of carbon per hectare (or 89 metric tons of carbon per acre) for the carbon stock density of U.S. forests in 2022, with an annual net change in carbon stock per area in 2022 of 0.67 metric tons of carbon sequestered per hectare per year (or 0.27 metric tons of carbon sequestered per acre per year).

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

Carbon Stock Density in Year 2022 = $(62,130 \text{ MMT C} \times 10^6) / (281,752 \text{ thou. hectares} \times 10^3) = \mathbf{221 \text{ metric tons of carbon stored per hectare}}$

Annual Net Change in Carbon Stock per Area in Year 2022 = $(-189 \text{ MMT C} \times 10^6) / (281,752 \text{ thou. hectares} \times 10^3) = \mathbf{-0.67 \text{ metric tons of carbon sequestered per hectare per year}^*}$

*Negative values indicate carbon sequestration.

From 2014 to 2022, the average annual sequestration of carbon per area was 0.58 metric tons C/hectare/year (or 0.23 metric tons C/acre/year) in the United States, with a minimum value of 0.52 metric tons C/hectare/year (or 0.21 metric tons C/acre/year) in 2019, and a maximum value of 0.67 metric tons C/hectare/year (or 0.27 metric tons C/acre/year) in 2022.

These values include carbon in the five forest pools: aboveground biomass, belowground biomass, dead wood, litter, and soil organic and mineral carbon, and are based on state-level Forest Inventory and Analysis (FIA) data. Forest carbon stocks and carbon stock change are based on the stock difference methodology and algorithms described by Smith, Heath, and Nichols (2010).

Conversion Factor for Carbon Sequestered in One Year by 1 Acre of Average U.S. Forest

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$-0.27 \text{ metric ton C/acre/year}^* \times (44 \text{ units CO}_2/12 \text{ units C}) = \textbf{-1.00 metric ton CO}_2\textbf{/acre/year sequestered annually by one acre of average U.S. forest.}$

*Negative values indicate carbon sequestration.

Please note that this is an estimate for “average” U.S. forests from 2021 to 2022; i.e., the annual net change in carbon stock for U.S. forests as a whole between 2021 and 2022. Significant geographical variations underlie the national estimates, and the values calculated here might not be representative of individual regions, states, or changes in the species composition of additional acres of forest.

To estimate carbon sequestered (in metric tons of CO₂) by additional "average" forestry acres in one year, multiply the number of additional acres by -1.0 metric ton CO₂ acre/year.

Sources

- EPA (2024). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Chapter 6 <https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-chapter-6-land-use-land-use-change-and-forestry_0.pdf>, Table 6-9: Net C Flux from Forest Ecosystem Pools in Forest Land Remaining Forest Land and Harvested Wood Pools (MMT C), Table 6-10: Forest Area (1,000 ha) and C Stocks in Forest Land Remaining Forest Land and Harvested Wood Pools (MMT C), Pg. 6-31, Pg. 6-31, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>)
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Acres of U.S. forest preserved from conversion to development

Forests are defined herein as managed forests that have been classified as forests for over 20 years (i.e., excluding forests converted to/from other land-use types). Please refer to the *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* for a discussion of the definition of U.S. forests and methodology for estimating carbon stored in U.S. forests (EPA 2024).

Based on data developed by the USDA Forest Service for the *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022*, the carbon stock density of U.S. forests in 2022 was 221 metric tons of carbon per hectare (or 89 metric tons of carbon per acre) (EPA 2024b). This estimate is composed of the five carbon pools: aboveground biomass (63 metric tons C/hectare), belowground biomass (11 metric tons C/hectare), dead wood (11 metric tons C/hectare), litter (13 metric tons C/hectare), and soil carbon, which includes mineral soils (101 metric tons C/hectare) and organic soils (21 metric tons C/hectare).

The *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* estimates soil carbon stock changes using U.S.-specific equations, IPCC guidance, and data from the USDA Natural Resource Inventory and the DayCent biogeochemical model (EPA 2024b). When calculating carbon stock changes in biomass due to conversion from forestland to development, the 2019 IPCC Refinement indicates that the average carbon stock change is equal to the carbon in biomass on the land use immediately after the conversion (e.g., development) minus the carbon lost that was in biomass on the land use prior to the conversion (e.g., forest land) plus the carbon stocks from one year of biomass growth and soil organic carbon stock change in the new or current land use (IPCC 2019). All of the biomass carbon from the prior land use is assumed to be lost in

the year the land use change occurs for conversion to development; therefore the carbon stock in annual development biomass after one year is conservatively estimated to be 0 metric tons C per hectare (IPCC 2019).

The averaged reference soil carbon stock (for high-activity clay, low-activity clay, sandy soils, and histosols for all climate regions in the United States) is 40.83 metric tons C/hectare (EPA 2024a). Carbon stock change in soils is time-dependent, with a default time period for transition between equilibrium soil carbon values of 20 years (IPCC 2006). Consequently, it is assumed that the change in equilibrium soil carbon will be annualized over 20 years to represent the annual flux in mineral and organic soils.

Organic soils also emit CO₂ when drained. Emissions from drained organic soils in forestland and drained organic soils in development vary based on the drainage depth and climate (IPCC 2006). The *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* estimates emissions from drained organic soils using U.S.-specific emission factors for development and IPCC (2014) default emission factors for forestland (EPA 2024a).

The annual change in emissions from one hectare of drained organic soils can be calculated as the difference between the emission factors for forest soils and development soils. The emission factors for drained organic soil on temperate forestland are 2.60 metric tons C/hectare/year and 0.31 metric tons C/hectare/year (EPA 2024a, IPCC 2014), and the average emission factor for drained organic soil on development for all climate regions (cool temperate, warm temperate, and subtropical) is 13.17 metric tons C/hectare/year (EPA 2024a). States identified as having drained organic soils include Florida, Georgia, Michigan, Minnesota, North Carolina, Virginia, Washington, and Wisconsin (EPA 2024b).

The IPCC (2006) Guidelines assume that all deadwood and litter carbon are lost in conversion to development and do not subsequently accumulate, hence the carbon stocks for these pools after conversion is assumed to be 0 (IPCC 2006).

Calculation for Converting U.S. Forests to U.S. Development

Annual Change in Biomass Carbon Stocks on Land Converted to Development Land-Use Category

$$\Delta \text{CB} = \Delta \text{C}_G + \text{C}_{\text{Conversion}} - \Delta \text{C}_L$$

Where:

ΔCB = annual change in carbon stocks in biomass on land converted to another land-use category (i.e., change in biomass on land converted from forest to development)

ΔC_G = annual increase in carbon stocks in biomass due to growth on land converted to another land-use category (i.e., 0.00 metric tons C/hectare on development one year after conversion from forestland)

$\text{C}_{\text{Conversion}}$ = initial change in carbon stocks in biomass on land converted to another land-use category. The sum of the carbon stocks in aboveground, belowground, dead wood, and litter biomass (-98.47 metric tons C/hectare). Immediately after conversion from forestland to development, the carbon stock of aboveground biomass is assumed to be zero, as the land is cleared of all vegetation for development)

ΔC_L = annual decrease in biomass stocks due to losses from harvesting, fuel wood gathering, and disturbances on land converted to another land-use category (assumed to be zero)

Therefore: $\Delta \text{CB} = \Delta \text{C}_G + \text{C}_{\text{Conversion}} - \Delta \text{C}_L = -98.47$ metric tons C/hectare/year of biomass carbon stocks are lost when forestland is converted to development in the year of conversion.

Annual Change in Organic Carbon Stocks in Mineral and Organic Soils

$$\Delta \text{C}_{\text{Soil}} = (\text{SOC}_0 - \text{SOC}_{(0-T)})/D$$

Where:

$\Delta \text{C}_{\text{Soil}}$ = annual change in carbon stocks in mineral and organic soils

SOC_0 = soil organic carbon stock in last year of inventory time period (i.e., 41.63 mt/hectare, the average reference soil carbon stock)

$\text{SOC}_{(0-T)}$ = soil organic carbon stock at beginning of inventory time period (i.e., 112 mt C/hectare, which includes 93 mt C/hectare in mineral soils plus 19 mt C/hectare in organic soils)

D = Time dependence of stock change factors which is the default time period for transition between equilibrium SOC values (i.e., 20 years)

Therefore: $\Delta C_{\text{Soil}} = (\text{SOC}_0 - \text{SOC}_{(0-T)})/D = (40.83 - 122)/20 = -4.06$ metric tons C/hectare/year of soil C lost.

Source: (IPCC 2006).

Annual Change in Emissions from Drained Organic Soils

The *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022* uses default IPCC (2014) factors for drained organic soil on forestland and U.S.-specific factors for development land. The change in emissions from drained organic soils per hectare is estimated as the difference between emission factors for drained organic forest soils and drained organic development soils.

$$\Delta L_{\text{Organic}} = \text{EF}_{\text{cropland}} - \text{EF}_{\text{forestland}}$$

Where:

$\Delta L_{\text{Organic}}$ = Annual change in emissions from drained organic soils per hectare

$\text{EF}_{\text{cropland}}$ = 13.17 metric tons C/hectare/year (average of emission factors for drained organic cropland soils in sub-tropical, cold temperate, and warm temperate climates in the United States) (EPA 2024a)

$\text{EF}_{\text{forestland}}$ = $2.60 + 0.31 = 2.91$ metric tons C/hectare/year (emission factors for temperate drained organic forest soils) (IPCC 2014)

$$\Delta L_{\text{Organic}} = 13.17 - 2.91 = 10.26 \text{ metric tons C/hectare/year emitted}$$

Consequently, the change in carbon density from converting forestland to development would be -98.47 metric tons of C/hectare/year of biomass plus -4.06 metric tons C/hectare/year of soil C, minus 10.26 metric tons C/hectare/year from drained organic soils, equaling a total loss of 112.79 metric tons C/hectare/year (or -45.64 metric tons C/acre/year) in the year of conversion. To convert to carbon dioxide, multiply by the ratio of the molecular weight of carbon dioxide to that of carbon (44/12), to yield a value of -413.56 metric tons CO₂/hectare/year (or -167.36 metric tons CO₂/acre/year) in the year of conversion.

Conversion Factor for Carbon Sequestered by 1 Acre of Forest Preserved from Conversion to Development

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$-45.64 \text{ metric tons C/acre/year}^* \times (44 \text{ units CO}_2/12 \text{ units C}) = \textbf{-167.36 metric tons CO}_2/\text{acre/year (in the year of conversion)}$

*Negative values indicate CO₂ that is NOT emitted.

To estimate CO₂ not emitted when an acre of forest is preserved from conversion to development, multiply the number of acres of forest not converted by -167.36 mt CO₂/acre/year. Note that this represents CO₂ avoided in the year of conversion. Please also note that this calculation method assumes that all of the forest biomass is oxidized during clearing (i.e., none of the burned biomass remains as charcoal or ash) and does not include any carbon stored in harvested wood products post-harvest. Also note that this estimate includes both mineral soil and organic soil carbon stocks.

Sources

- EPA (2024a). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Annex 3b <<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-annex-3-additional-source-or-sink-categories-part-b.pdf>>, Table A-179: U.S. Soil Groupings Based on the IPCC Categories and Dominant Taxonomic Soil, and Reference Carbon Stocks (Metric Tons C/ha), Pg. A-374
- EPA (2024b). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Chapter 6 <https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-chapter-6-land-use-land-use-change-and-forestry_0.pdf>, Table 6-10: Forest Area (1,000 ha) and C Stocks in Forest Land Remaining Forest Land and Harvested Wood Pools (MMT C), Pg. 6-31, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>)
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- IPCC (2019). 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 (Agriculture, Forestry and Other Land Use), Chapter 5, Table 5.9. Intergovernmental Panel on Climate Change, Geneva, Switzerland [✉ <https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol4.html>](https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol4.html).

Propane cylinders used for home barbecues

Propane is 81.8 percent carbon (EPA 2024). The fraction oxidized is assumed to be 100 percent (IPCC 2006).

Carbon dioxide emissions per pound of propane were determined by multiplying the weight of propane in a cylinder times the carbon content percentage times the fraction oxidized times the ratio of the molecular weight of carbon dioxide to that of carbon (44/12). Propane cylinders vary with respect to size; for the purpose of this equivalency calculation, a typical cylinder for home use was assumed to contain 16 pounds of propane.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

16 pounds propane/1 cylinder × 0.818 pounds C/pound propane × 0.4536 kilograms/pound × 44 kg CO₂/12 kg C × 1 metric ton/1,000 kg = **0.022 metric tons CO₂/cylinder**

Sources

- EPA (2024). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Annex 2 (Methodology for estimating CO₂ emissions from fossil fuel combustion) <<https://epa.gov/system/files/documents/2024-04/us-ghg-inventory-2024-annex-2-emissions-fossil-fuel-combustion.pdf>>, Table A-32 " Physical Characteristics of Hydrocarbon Gas Liquids," pg. A-78, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (790 pp, 15.9 MB, About PDF <<https://epa.gov/home/pdf-files>>).
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Railcars of coal burned

The average heat content of coal consumed by the electric power sector in the U.S. in 2022 was 20.71 mmbtu per metric ton (EIA 2024). The average carbon coefficient of coal combusted for electricity generation in 2022 was 26.13 kilograms carbon per mmbtu (EPA 2024). The fraction oxidized is assumed to be 100 percent (IPCC 2006).

Carbon dioxide emissions per ton of coal were determined by multiplying heat content times the carbon coefficient times the fraction oxidized times the ratio of the molecular weight of carbon dioxide to that of carbon (44/12). The amount of coal in an average railcar was assumed to be 100.19 short tons, or 90.89 metric tons (Hancock 2001).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$20.71 \text{ mmbtu/metric ton coal} \times 26.13 \text{ kg C/mmbtu} \times 44 \text{ kg CO}_2/12 \text{ kg C} \times 90.89 \text{ metric tons coal/railcar} \times 1 \text{ metric ton/1,000 kg} = \mathbf{180.4 \text{ metric tons CO}_2/\text{railcar}}$

Sources

- EIA (2024). Monthly Energy Review May 2024, Table 5: Approximate Heat Content of Coal and Coal Coke [🔗](https://www.eia.gov/totalenergy/data/monthly/pdf/mer.pdf) <<https://www.eia.gov/totalenergy/data/monthly/pdf/mer.pdf>>. (PDF) (272 pp, 3.04 MB, About PDF <<https://epa.gov/home/pdf-files>>)

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<<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>>.

Pounds of coal burned

The average heat content of coal consumed by the electric power sector in the U.S. in 2022 was 20.71 mmbtu per metric ton (EIA 2024). The average carbon coefficient of coal combusted for electricity generation in 2022 was 26.13 kilograms carbon per mmbtu (EPA 2024). The fraction oxidized is 100 percent (IPCC 2006).

Carbon dioxide emissions per pound of coal were determined by multiplying heat content times the carbon coefficient times the fraction oxidized times the ratio of the molecular weight of carbon dioxide to that of carbon (44/12).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

20.71 mmbtu/metric ton coal × 26.13 kg C/mmbtu × 44 kg CO₂/12 kg C × 1 metric ton coal/2,204.6 pound of coal × 1 metric ton/1,000 kg = **9.00 x 10⁻⁴ metric tons CO₂/pound of coal**

Sources

- EIA (2024). Monthly Energy Review May 2024, Table 5: Approximate Heat Content of Coal and Coal Coke [↗](https://www.eia.gov/totalenergy/data/monthly/pdf/mer.pdf) <<https://www.eia.gov/totalenergy/data/monthly/pdf/mer.pdf>>. (PDF) (272 pp, 3.04 MB, About PDF <<https://epa.gov/home/pdf-files>>)
- EPA (2024). Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022. Annex 2 (Methodology for estimating CO₂ emissions from fossil fuel combustion), Table A-20 "C Content Coefficients by Year (MMT C/QBtu)" pg A-51, U.S. Environmental Protection Agency, Washington, DC. U.S. EPA #430-R-24-004 (PDF) (96 pp, 3 MB, About PDF <<https://epa.gov/home/pdf-files>>).
- IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2 (Energy). Intergovernmental Panel on Climate Change, Geneva, Switzerland [↗](https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html) <<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>>.

Tons of waste recycled instead of landfilled

To develop the conversion factor for recycling rather than landfilling waste, emission factors from EPA's Waste Reduction Model (WARM) were used (EPA 2024). These emission factors were developed following a life-cycle assessment methodology using estimation techniques developed for national inventories of greenhouse gas emissions. According to WARM, the net emission reduction from recycling mixed recyclables (e.g., paper, metals, plastics), compared with a baseline in which the materials are landfilled (i.e., accounting for the avoided emissions from landfilling), is 2.83 metric tons of carbon dioxide equivalent per short ton.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

2.83 metric tons CO₂ equivalent/ton of waste recycled instead of landfilled

Sources

- EPA (2024). Waste Reduction Model (WARM), Version 16. U.S. Environmental Protection Agency <<https://epa.gov/warm/versions-waste-reduction-model-warm#16>>.

Number of garbage trucks of waste recycled instead of landfilled

The carbon dioxide equivalent emissions avoided from recycling instead of landfilling 1 ton of waste are 2.83 metric tons CO₂ equivalent per ton, as calculated in the “Tons of waste recycled instead of landfilled” section above.

Carbon dioxide emissions reduced per garbage truck full of waste were determined by multiplying emissions avoided from recycling instead of landfilling 1 ton of waste by the amount of waste in an average garbage truck. The amount of waste in an average garbage truck was assumed to be 7 tons (EPA 2002).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

2.83 metric tons CO₂ equivalent /ton of waste recycled instead of landfilled x 7 tons/garbage truck = **19.81 metric tons CO₂e/garbage truck of waste recycled instead of landfilled**

Sources

- EPA (2024). Waste Reduction Model (WARM), Version 16. U.S. Environmental Protection Agency <<https://epa.gov/warm/versions-waste-reduction-model-warm#16>>.
- EPA (2002). Waste Transfer Stations: A Manual for Decision-Making. U.S. Environmental Protection Agency (PDF) <<https://epa.gov/sites/production/files/2016-03/documents/r02002.pdf>> (66 pp, 523 KB, About PDF <<https://epa.gov/epahome/pdf.html>>).

Trash bags of waste recycled instead of landfilled

According to WARM, the net emission reduction from recycling mixed recyclables (e.g., paper, metals, plastics), compared with a baseline in which the materials are landfilled (i.e., accounting for the avoided emissions from landfilling), is 2.83 metric tons CO₂ equivalent per short ton, as calculated in the “Tons of waste recycled instead of landfilled” section above.

Carbon dioxide emissions reduced per trash bag full of waste were determined by multiplying emissions avoided from recycling instead of landfilling 1 ton of waste by the amount of waste in an average trash bag.

The amount of waste in an average trash bag was calculated by multiplying the average density of mixed recyclables by the average volume of a trash bag.

According to EPA’s standard volume-to-weight conversion factors, the average density of mixed recyclables is 111 lbs per cubic yard (EPA 2016a). The volume of a standard sized household trash bag in the U.S. was assumed to be 13 gallons, based on the typical size of household kitchen trash cans.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$$2.83 \text{ metric tons CO}_2 \text{ equivalent /short ton of waste recycled instead of landfilled} \times 1 \text{ short ton}/2,000 \text{ lbs} \times 111 \text{ lbs of mixed recyclables/cubic yard} \times 1 \text{ cubic yard}/173.57 \text{ dry gallons} \times 13 \text{ gallons/trash bag} = \mathbf{1.18 \times 10^{-2} \text{ metric tons CO}_2 \text{ equivalent/trash bag of waste recycled instead of landfilled}}$$

Sources

- EPA (2024). Waste Reduction Model (WARM), Version 15.2. U.S. Environmental Protection Agency <<https://epa.gov/warm/versions-waste-reduction-model-warm#16>>.

- EPA (2016a). Volume-to-Weight Conversion Factors (PDF) <https://epa.gov/sites/production/files/2016-04/documents/volume_to_weight_conversion_factors_memorandum_04192016_508fml.pdf>. Office of Resource Conservation and Recovery. April 2016. U.S. Environmental Protection Agency. (7 pp, 318 KB, About PDF <<https://epa.gov/home/pdf-files>>)

Coal-fired power plant emissions for one year

In 2022, a total of 197 power plants used coal to generate at least 95% of their electricity (EPA 2024). These plants emitted 746,630,725.6 metric tons of CO₂ in 2022.

Carbon dioxide emissions per power plant were calculated by dividing the total emissions from power plants whose primary source of fuel was coal by the number of power plants.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$746,630,725.6 \text{ metric tons of CO}_2 \times 1/197 \text{ power plants} = \mathbf{3,790,003.72 \text{ metric tons CO}_2/\text{power plant}}$

Sources

- EPA (2024). eGRID <<https://epa.gov/egrid>> year 2022 data. U.S. Environmental Protection Agency, Washington, DC.

Natural gas-fired power plant emissions for one year

In 2022, a total of 1,637 power plants used natural gas to generate at least 95% of their electricity (EPA 2024). These plants emitted 625,669,621.5 metric tons of CO₂ in 2022.

Carbon dioxide emissions per power plant were calculated by dividing the total emissions from power plants whose primary source of fuel was natural gas by the number of power plants.

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$625,669,621.5 \text{ metric tons of CO}_2 \times 1/1,637 \text{ power plants} = \mathbf{382,205.02 \text{ metric tons CO}_2/\text{power plant}}$

Sources

- EPA (2024). eGRID <<https://epa.gov/egrid>> year 2022 data. U.S. Environmental Protection Agency, Washington, DC.

Number of wind turbines running for a year

In 2023, the average nameplate capacity of wind turbines installed in the U.S. was 1.76 MW (Hoen et al. 2023). The average wind capacity factor in the U.S. in 2022 was 36 percent (DOE 2023).

Electricity generation from an average wind turbine was determined by multiplying the average nameplate capacity of a wind turbine in the United States (1.76 MW) by the average U.S. wind capacity factor (0.36) and by the number of hours per year. It was assumed that the electricity generated from an installed wind turbine would replace marginal sources of grid electricity.

The U.S. annual wind national marginal emission rate to convert reductions of kilowatt-hours into avoided units of carbon dioxide emissions is 6.03×10^{-4} (EPA 2024).

Carbon dioxide emissions avoided per year per wind turbine installed were determined by multiplying the average electricity generated per wind turbine in a year by the annual wind national marginal emission rate (EPA 2024).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$1.76 \text{ MW average capacity} \times 0.36 \times 8,760 \text{ hours/year} \times 1,000 \text{ kWh/MWh} \times 6.0273 \times 10^{-4}$
metric tons CO₂/kWh reduced = **3,348 metric tons CO₂/year/wind turbine installed**

Sources

- DOE (2023). Wind Market Reports: 2022 Edition [🔗](https://www.energy.gov/eere/wind/wind-market-reports-2023-edition) <<https://www.energy.gov/eere/wind/wind-market-reports-2023-edition>>. U.S. Department of Energy, Washington, DC.
- Hoen, B.D., Diffendorfer, J.E., Rand, J.T., Kramer, L.A., Garrity, C.P., and Hunt, H.E., 2018, United States Wind Turbine Database v5.3 (January 13, 2023): U.S. Geological Survey, American Clean Power Association, and Lawrence Berkeley National Laboratory data release, <https://doi.org/10.5066/F7TX3DN0> [🔗](https://doi.org/10.5066/f7tx3dn0) <<https://doi.org/10.5066/f7tx3dn0>>.
- EPA (2024) AVERT <<https://epa.gov/statelocalenergy/download-avert>>, U.S. annual wind national marginal emission rate, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.

Number of smartphones charged

According to U.S. DOE, the 24-hour energy consumed by a common smartphone battery is 28.446 Watt-hours (Dommu 2024). This includes the amount of energy needed to charge a fully depleted smartphone battery and maintain that full charge throughout the day. The average time required to completely recharge a smartphone battery is 2 hours (Ferreira et al. 2011). Maintenance mode power, also known as the power consumed when the phone is fully charged and the charger is still plugged in, is 0.411 Watts (Dommu 2024). To obtain the amount of energy consumed to charge the smartphone, subtract the amount of energy consumed in “maintenance mode” (0.411 Watts times 22 hours) from the 24-hour energy consumed (28.446 Watt-hours).

Carbon dioxide emissions per smartphone charged were determined by multiplying the energy use per smartphone charged by the national weighted average carbon dioxide emission rate for delivered electricity. The national weighted average carbon dioxide

emission rate for delivered electricity in 2022 was 1,405.3 lbs CO₂ per megawatt-hour, which accounts for losses during transmission and distribution (EPA 2024).

Calculation

Note: Due to rounding, performing the calculations given in the equations below may not return the exact results shown.

$[28.446 \text{ Wh} - (22 \text{ hours} \times 0.411 \text{ Watts})] \times 1 \text{ kWh}/1,000 \text{ Wh} = 0.019 \text{ kWh/smartphone charged}$

$0.019 \text{ kWh/charge} \times 1,405.3 \text{ pounds CO}_2/\text{MWh delivered electricity} \times 1 \text{ MWh}/1,000 \text{ kWh} \times 1 \text{ metric ton}/2,204.6 \text{ lbs} = \mathbf{1.24 \times 10^{-5} \text{ metric tons CO}_2/\text{smartphone charged}}$

Sources

- Jeremy Dommu, U.S. DOE, personal communication, October 4th, 2024.
- EPA (2024a). eGRID <<https://epa.gov/egrid>>. U.S. marginal and baseload emission rates, year 2022 data. U.S. Environmental Protection Agency, Washington, DC.
- Federal Register (2016). Energy Conservation Program: Energy Conservation Standards for Battery Chargers; Final Rule, page 38,284 (PDF) [↗](https://www.gpo.gov/fdsys/pkg/fr-2016-06-13/pdf/2016-12835.pdf) <<https://www.gpo.gov/fdsys/pkg/fr-2016-06-13/pdf/2016-12835.pdf>> (71 pp, 0.7 MB, About PDF <<https://epa.gov/home/pdf-files>>).
- Ferreira, D., Dey, A. K., & Kostakos, V. (2011). Understanding Human-Smartphone Concerns: A Study of Battery Life. Pervasive Computing, pp.19-33. doi:10.1007/978-3-642-21726-5_2 [↗](https://doi.org/10.1007/978-3-642-21726-5_2) <https://doi.org/10.1007/978-3-642-21726-5_2>.

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